Towards Implicit Personal Eye Gaze Calibration in Real-world Driving Scenarios

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Abstract

Various studies have been approached in remote eye gaze monitoring technologies to estimate drivers' eye gaze with high user-friendliness. However, current methods require a tedious calibration process or specific hardware and have poor performance on uncalibrated drivers. In this paper, an implicit gaze calibration method is presented for in-vehicle driver gaze prediction systems under real-world driving scenarios. Considering the topological position and head motion characteristics, it locates the representative gaze zones by clustering, and progressively selects the head dynamics between the target coarse gaze zones. Benefiting from the linear interpolation, the gaze points of the candidate head dynamics are refined to their vertical projection trajectory. Both the representative gaze zones and clustered gaze points on the head dynamics can be utilized as calibration points. Extensive experiment results demonstrate that the proposed method is effective and comparable with manual gaze calibration.

ACM Reference Format:
Yafei Wang, Guoliang Yuan, and Xianping Fu. 2025. Towards Implicit Personal Eye Gaze Calibration in Real-world Driving Scenarios. In *2025 Symposium on Eye Tracking Research and Applications (ETRA '25), May 26–29, 2025, Tokyo, Japan.* ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/3715669.3726793

1 Introduction

The driver's eye gaze is strongly related to their driving actions, and has been regarded as a crucial indicator of driver's visual attention [Wang et al. 2021]. Gaze calibration is the particular step in eye gaze estimation to establish the mapping function from the driver's eye features to the gaze direction. Manual gaze calibration methods [Fu et al. 2012; Yamashiro et al. 2009; Yang et al. 2019] usually require a tedious and time-consuming recalibration, making the gaze tracking system person-exclusive and session-exclusive. Several gaze calibration methods were designed for the driving condition by exploring the driving gaze behavior [Yuan et al. 2022] and gaze frequency [Fu et al. 2012]. These methods are heavily dependent on the given calibration points and cannot generalize and apply to surrounding gaze points with high accuracy. Since the calibrated gaze points are sparse in the field of view, these methods still cannot meet the quantity requirements under large head rotation conditions.

This paper wants to further explore two issues: one is how to find effective gaze calibration points; the other is how to improve the accuracy and generalization ability of driver gaze zone classification and gaze estimation by adding calibration points. To address this, an implicit gaze calibration method *CalibDriver* is proposed for automatic driver gaze calibration in this paper. This method fully considers the topological positional relationship and head motion characteristics between head dynamics and coarse gaze zones during driving. It finds the effective gaze points in the head dynamics between the forward-view gaze zone and the target gaze zones, and increases the available range and gaze points for the driver gaze mapping. Despite being an unsupervised method, the proposed method is comparable with the manual gaze calibration method in terms of gaze prediction.

2 Proposed Method

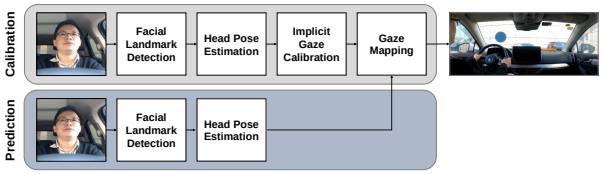

Figure 1: The overall framework of the driving attention monitoring system.

The overall framework of the driving attention monitoring system is shown in Fig 1. In the driving attention monitoring system, the driver's facial landmarks and head pose are inferred from the face camera by facial landmark detection and head pose estimation. During the gaze calibration step, automatic gaze mapping is achieved by implicit gaze calibration.

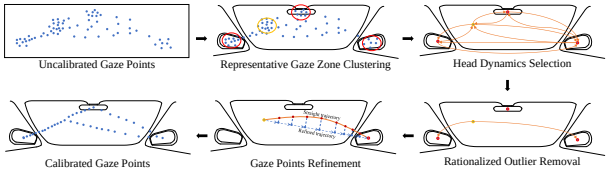

Figure 2: Illustration of the *CalibDriver* method.

Our primary goal is to find the valid gaze calibration points in an unsupervised manner. The driver's facial landmarks and head

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ETRA '25, Tokyo, Japan
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ACM ISBN 979-8-4007-1487-0/25/05
https://doi.org/10.1145/3715669.3726793

pose are utilized as driver features to construct the uncalibrated gaze points. Representative gaze zones are first extracted by head pose-relevant clustering. Through head dynamics learning, valid gaze point pairs then can be found for gaze mapping model training, in which the available gaze points are located on the head dynamics trajectories. Supporting by the rationalized outlier removal and gaze points refinement, the valid gaze calibration points on the head dynamic trajectories are found. Thus, the mapping model between driver features and gaze points is established for automatic driver gaze estimation. The illustration of the implicit gaze calibration method is shown in Fig 2.

3 Experimental Results

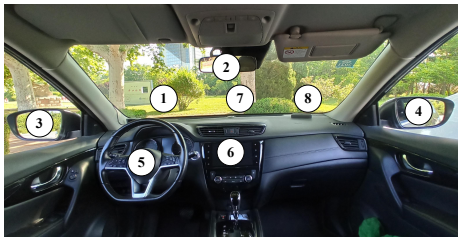

Figure 3: Illustration of gaze zone partition.

Dataset. To collect the driver's face video in naturalistic condition, one camera is mounted with bracket above the dashboard and in front of the windshield, and its front camera can acquire the driver's face video in full view. The coarse gaze is partitioned into eight gaze zones, as shown in Fig. 3. The dataset contains two subset, Set#1 manual calibration dataset and Set#2 continuous calibration data. In the Set#1 manual calibration data, the driver was instructed to turn their heads to face the marker of each gaze zone individually. The training data is 2,400 frames, the validation data is 800 frames. In the Set#2 continuous calibration data, the driver was asked to simulate the head dynamics between forward gaze zone and other gaze zones. After removing the data without head pose estimation results, the total valid data is 6,000 frames. This data is used for calibration model training. All methods are tested on Set#1 validation data.

Features. The head pose, eye pose, and facial landmarks are extracted as driver features for calibration model training. All features are normalized and smoothed by the Gaussian filter of size 40. The feature dimension of head pose, eye pose, and facial landmarks is 6, 6, and 136, respectively.

Pre-trained gaze mapping. In this paper, head pose angle values are applied as pre-trained gaze mapping model. It means that the head pose angle values are the uncalibrated gaze points. Typically, head pose can determine most of the gaze direction. head pose angle values are also utilized to head dynamics.

Clustering. The clustering of candidate gaze zones is modelled by Gaussian Mixture Model, and solved by Expectation-Maximization algorithm. The parameters are set as with default values. The clustering center is treated as the gaze zone center.

Calibration. The two-dimensional single Gaussian Process Regression model is used for both manual calibration methods and the proposed implicit method. The model is with default parameters. For each method, the model is retrained with given data and labels.

Results. The comparison results with manual calibration methods are shown in Table. 1. The gaze prediction error is calculated on the mapping scene image in the x direction and y direction. The success detection rate of the gaze zone (SDR) is computed based on the nearest neighbor classification, which can reveal the precision of the gaze prediction. G3, G4, G5, and G6 denote that methods using data from gaze zone 1-3, gaze zone 1-4, gaze zone 1-5, and gaze zone 1-6, respectively. *CalibDriver*-G4 and *CalibDriver*-G6 denote proposed method generates four and six representative gaze zone with head dynamics, respectively.

Table 1: Comparisons with manual calibration methods.

Methods	Gaze Error		
	X (pixel)	Y (pixel)	SDR (%)
G3 Manual Calibration	99.8002	128.9408	50.00
G4 Manual Calibration	52.9999	128.0858	62.50
G5 Manual Calibration	41.3174	74.1205	75.00
G6 Manual Calibration	32.7151	24.0789	87.50
CalibDriver-G4	46.7699	30.4049	100.00
CalibDriver-G6	38.2947	21.4824	100.00

4 Conclusion

In this paper, an implicit gaze calibration method is proposed for in-vehicle driver gaze tracking systems in an unsupervised manner. The proposed method is an implicit calibration method, and without any manual cooperation. The experimental results demonstrate the effectiveness of the proposed method under cross-session condition.

Acknowledgments

This work was supported in part by the National Key Research and Development Program of China Grant 2022YFE0132600, by the National Natural Science Foundation of China Grants 62002043, by the Research Project of China Disabled Persons' Federation - on Assistive Technology Grant 2024CDPFAT-28.

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